MBA · DIGITAL BUSINESS VALUATION

Week 4: Relative Valuation

Multiples, fundamentals and applications in Indonesia

Based on Aswath Damodaran's Investment Valuation, Chapters 17–20
and the supplied Sessions 14–18



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Learning Outcomes

1 Explain relative value

Compare market pricing while recognizing the risk of sector-wide mispricing.

2 Define consistent multiples

Match equity and enterprise measures, accounting policies and time periods.

3 Select comparable firms

Evaluate business economics, growth, risk and cash conversion.

4 Interpret key drivers

Explain how ROE, margins and reinvestment affect valuation multiples.

5 Estimate a value range

Normalize data, test sensitivities and reconcile enterprise and equity value.

6 Value digital businesses

Distinguish GMV from revenue and assess the path to sustainable margins.

Roadmap

1	**Foundations**	Definitions, peer selection and distribution
2	**P/E ratios**	Earnings quality, growth, risk and ROE
3	**Enterprise multiples**	EV/EBITDA, EV/EBIT and the equity bridge
4	**Book value multiples**	P/BV, ROE and the economics of banking
5	**Revenue multiples**	Margins, reinvestment and digital platforms
6	**Application workshop**	Indonesian cases, sensitivities and DCF cross-checks

Relative valuation and intrinsic valuation

Dimension	Relative valuation	Intrinsic valuation
Core question	How do comparable assets trade?	What are future cash flows worth? Risk : CF uncertainty, Assumption of growth, Assumption of Ke,
Anchor	Observed market multiples	Cash flows and required returns or cost
Main assumption	Peers provide a useful pricing benchmark	Cash-flow and risk forecasts are credible FORECAST : Q and Price, Demand
Main vulnerability	Shared sector mispricing euphoria, sometimes undervalue/overvalued	Forecast and terminal-value errors

A company can look inexpensive relative to peers and still trade above intrinsic value.

A lower share price does not mean a cheaper stock

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Hypothetical stock	Price per share real market price	EPS	P/E Value
Nusantara A	IDR 500	IDR 25	20.0×
Nusantara B	IDR 2,000	IDR 200	10.0× this is cheaper

Standardized price = market value ÷ a relevant operating or equity measure

value = how they create earning

B has the lower P/E. Its growth, risk and earnings quality still determine whether it offers value.

Four tests for every multiple

1 Define

Who owns the numerator and denominator? Which period and accounting basis apply?

2 Describe

What do the median, dispersion, outliers and missing observations show?

3 Analyze

Which fundamentals explain the multiple and how do they interact?

4 Apply

Which peers are credible and what adjustments does the target require?

Equity and enterprise multiples

Claimholder basis	Consistent multiples	Key denominator
Common equity	P/E income	Earnings attributable to common equity
Common equity	P/BV asset	Common shareholders' book equity
Operating enterprise	EV/EBITDA and EV/EBIT	Consolidated operating earnings
Operating enterprise	EV/Sales	Consolidated operating revenue
Operating enterprise	EV/Invested capital	Capital invested in operating assets

EV/net income mixes claimholders. P/S is widely used, but leverage complicates its interpretation.

Enterprise value and the equity bridge

EV = Equity value + Debt + Preferred equity + NCI
− Cash − Non-operating investments

NCI = non-controlling interests. Define which cash and investments fall outside the operating valuation.

Equity value = EV − Debt − Preferred equity − NCI
+ Cash + Non-operating investments

Match consolidation scope, valuation date and lease treatment across every peer.

gold company: jewelry, gold investment

Comparable firms share economic characteristics

Screen	Evidence to compare
Business model	Revenue source, customer mix, geography and segment exposure
Growth and profitability	Expected growth, margins, ROE or ROIC and maturity
Risk and funding	Leverage, earnings volatility, country exposure and liquidity
Cash conversion	Capital expenditure, working capital and recurring reinvestment
Measurement quality	Comparable accounting, disclosure dates and denominator definitions

A sector label is a starting point. A diversified business may require segment-level peers.

One date and one denominator convention

Market values at a common date

Use contemporaneous prices and current shares outstanding for market capitalization.

A consistent earnings horizon

Distinguish last fiscal year, trailing 12 months, next fiscal year and next 12 months.

A consistent ownership basis

Use attributable common earnings and compatible dilution conventions for per-share measures.

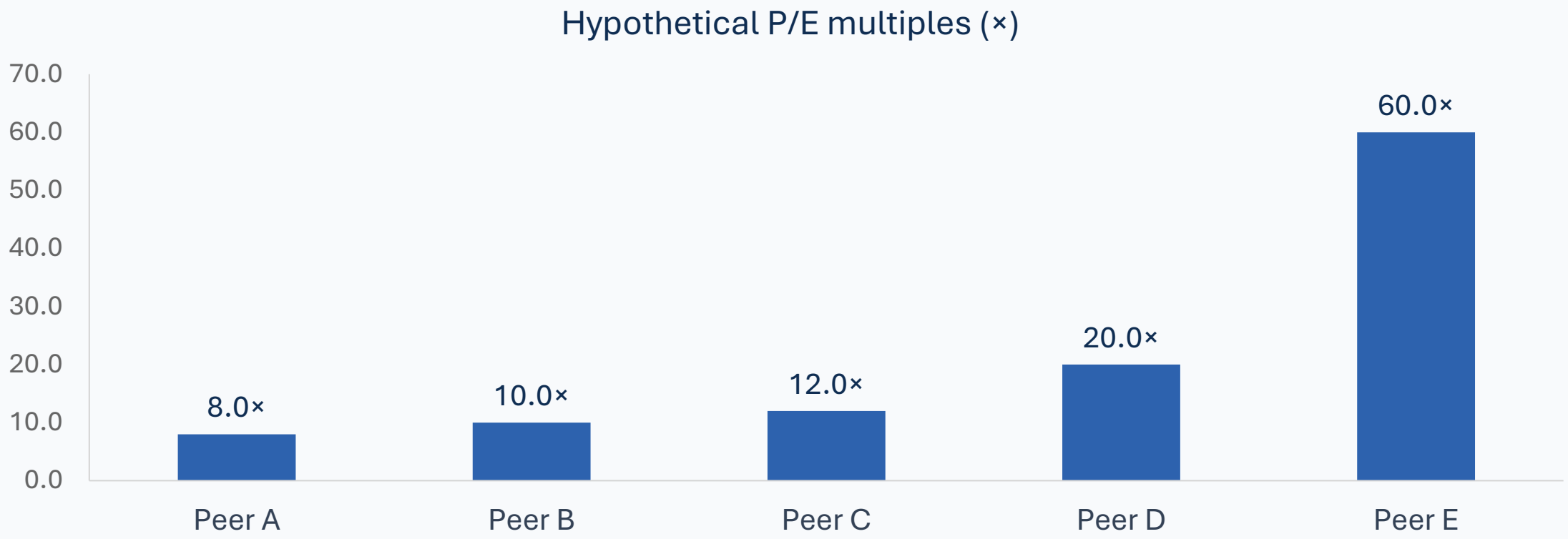
A documented adjustment log

Reconcile reported and normalized results, including taxes, disposals and accounting differences.

DISTRIBUTION

Outliers can change the apparent benchmark

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Mean 22.0× vs median 12.0×. Explain the 60× observation and disclose loss-making exclusions.

APPLICATION

Peer adjustments require evidence

Narrow peer sets

Useful for close matches, but small samples may be unstable or selective.

Fundamental adjustments

Explain a premium or discount through growth, risk, returns and cash conversion.

Cross-sectional regression

Estimate a conditional benchmark only with enough credible observations and sensible variables.

Residuals as research leads

A negative pricing residual may reflect omitted risk, weak data or genuine mispricing.

P/E: trailing and forward earnings

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Trailing P/E = $Price_0$ / EPS of the last 12 months
Forward P/E = $Price_0$ / Expected EPS_1

Label the forecast horizon. “Forward” can mean next fiscal year or next 12 months.

Illustrative inputs	Trailing	Forward
Price = IDR 1,500	EPS = IDR 100	EPS = IDR 125
Result	15.0×	12.0×

The lower forward P/E reflects expected earnings growth. It does not establish undervaluation.

P/E embeds growth, risk and reinvestment

Forward P/E = Payout ratio / (k_e – g)
Trailing P/E = Payout ratio × (1 + g) / (k_e – g)

k_e = cost of equity. g = sustainable growth. Under g = retention × ROE, payout = 1 – g/ROE.

Stable-growth model with sustainable payout and $g < k_e$.

Forward P/E = (1 – g/ROE) / (k_e – g)

Higher growth requires reinvestment. Its value depends on the return earned on that reinvestment.

A higher ROE can support a higher P/E

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Assumption	High-return firm	Low-return firm
Sustainable ROE	20%	8%
Growth / cost of equity	4% / 12%	4% / 12%
Retention required: g / ROE	20%	50%
Sustainable payout	80%	50%
Justified forward P/E	10.00×	6.25×

Identical growth can imply different value because one firm must reinvest more to achieve it.

Earnings normalization changes the conclusion

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Earnings reconciliation	IDR billion
Reported net income	600
Less: after-tax gain on asset sale	(150)
Add: after-tax exceptional restructuring cost	50
Normalized net income	500
Market capitalization	7,500
Reported P/E / normalized P/E	12.5× / 15.0×

An adjustment is credible only if its economic rationale is consistent across target and peers.

Indonesian consumer case: a P/E range

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Fictional peer	Forward P/E	Growth	ROE
Pangan A	14.0×	7%	16%
Pangan B	16.0×	9%	19%
Pangan C	18.0×	11%	23%
Target	To estimate	8%	18%

Target forward net income: IDR 500 billion. Shares: 5 billion.

Selected teaching range: 14–16×

Equity value = IDR 7.0–8.0 trillion

Value per share = IDR 1,400–1,600

The range is an analyst judgment, conditional on similar risk and earnings quality.

Low P/E and low PEG can be misleading

Cyclical peak earnings

A temporary profit surge lowers P/E just when normalized earnings may be much lower.

Negative or near-zero earnings

P/E becomes uninformative or unstable. An alternative denominator needs its own economic test.

PEG = P/E ÷ expected EPS growth in percentage points

15× / 10 = 1.5. This shortcut does not remove differences in risk, ROE or growth duration.

Leverage and buybacks

Financing changes may alter EPS and equity risk without improving operating economics.

EV/EBITDA and EV/EBIT measure different things

Multiple	Useful interpretation	Important limitation
EV/EBITDA	Value per unit of operating earnings before D&A	Ignores taxes, capex and working capital
EV/EBIT	Value relative to earnings after D&A	Depreciation policies and asset age matter
EV/FCFF	Value relative to cash available to capital providers	Annual reinvestment can make FCFF volatile

Similar EV/EBITDA does not imply similar cash-flow value when capital intensity differs.

EBITDA is only the starting point for cash flow

$$FCFF = EBIT \times (1 - t) + D\&A - Capex - \Delta WC$$

Substituting EBIT = EBITDA − D&A gives:

$$FCFF = EBITDA \times (1 - t) + D\&A \times t - Capex - \Delta WC$$

Cash taxes and reinvestment determine how much EBITDA can support enterprise value.

Justified EV/EBITDA follows cash conversion

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$EV / EBITDA_1 = (FCFF_1 / EBITDA_1) / (WACC - g)$

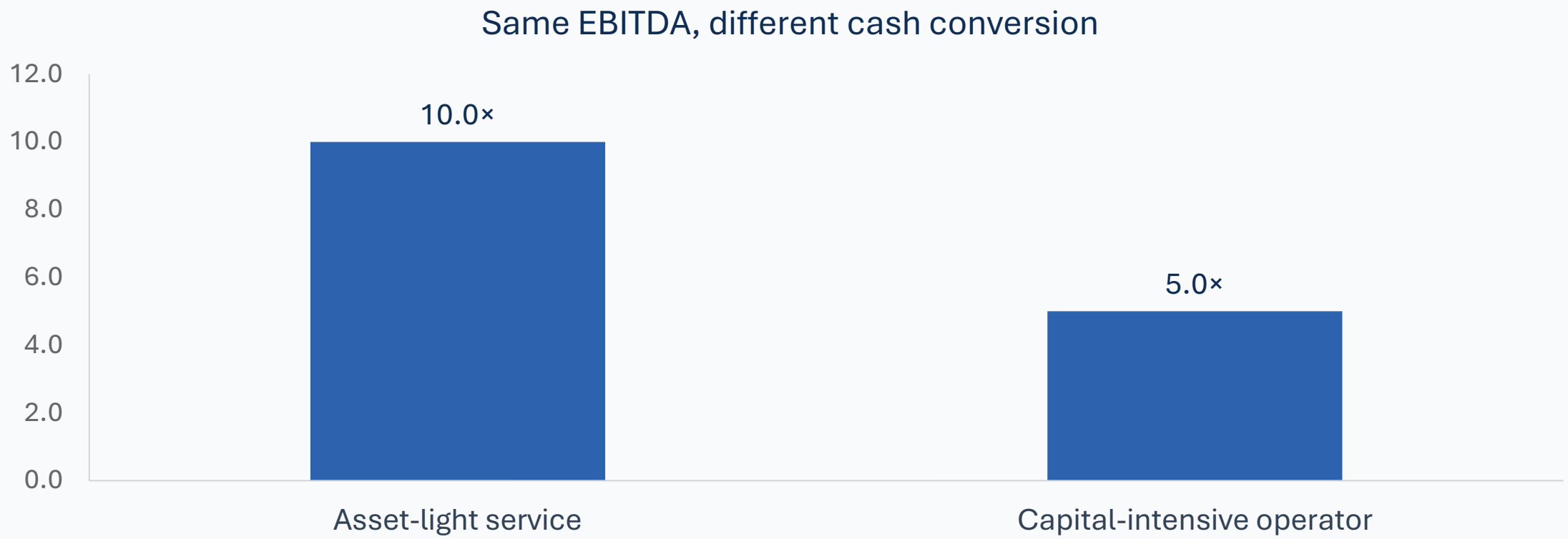
Stable growth, WACC > g, and sustainable operating and reinvestment ratios.

Illustrative forward assumptions	Value
Tax / D&A-to-EBITDA	25% / 20%
Capex / ΔWC as % of EBITDA	30% / 5%
FCFF-to-EBITDA	75% + 5% − 30% − 5% = 45%
WACC / growth / justified multiple	10% / 4% / 7.5×

ENTERPRISE APPLICATION

Equal EBITDA can produce unequal value

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FCFF/EBITDA of 60% vs 30%, with WACC 10% and growth 4%, implies 10× vs 5×.

ENTERPRISE APPLICATION

Indonesian infrastructure case: enterprise to equity

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Valuation bridge	IDR billion
Forward EBITDA × selected multiple	1,200 × 8.0×
Enterprise value	9,600
Less: debt / non-controlling interests	(3,000) / (400)
Add: cash outside operating value	500
Common equity value	6,700
5 billion shares: value per share	IDR 1,340

Ignoring NCI would overstate common equity value when EBITDA consolidates the full subsidiary.

ENTERPRISE APPLICATION

Enterprise multiple comparability traps

Lease accounting

Include lease liabilities and use an EBITDA definition consistent with all selected peers.

Consolidation and associates

Match subsidiary earnings to NCI and separate investments whose earnings are outside EBITDA.

Adjusted EBITDA

Repeated restructuring and share-based compensation can represent real economic costs.

Capex and cyclicality

Compare maintenance needs and normalized earnings across the business cycle.

P/BV connects market price with book equity

P/BV = Market value of common equity
/ Book value of common equity

Book equity is an accounting measure. It is not automatically liquidation value or replacement cost.

EV / Invested capital = Operating enterprise value
/ Operating invested capital

Use a consistent treatment of cash, debt, goodwill and capitalized intangible investment.

P/BV depends on the spread between ROE and k_e

Justified P/BV = (ROE − g) / (k_e − g)

ROE uses next-period earnings over opening book equity. Growth follows retention × ROE.

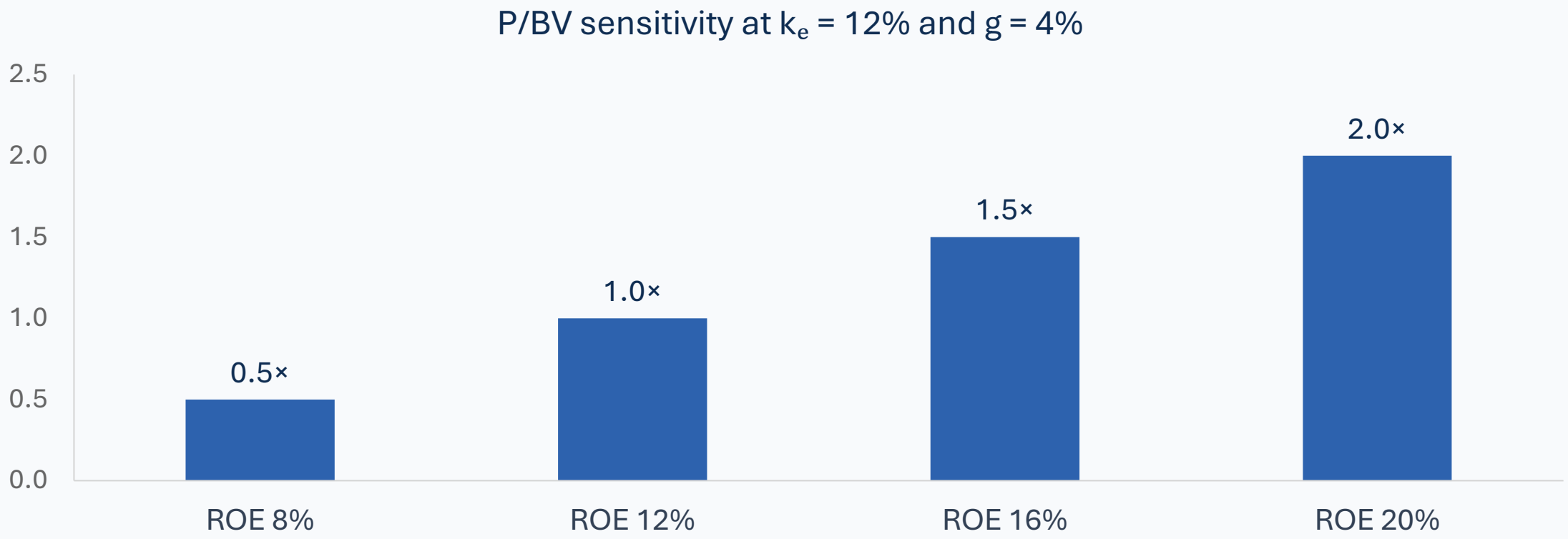
Stable growth, clean-surplus consistency and g < k_e.

Sustainable economics	Implication
ROE > cost of equity	P/BV > 1 under the model
ROE = cost of equity	P/BV = 1
ROE < cost of equity	P/BV < 1 can be economically justified

A discount to book value alone is insufficient evidence of undervaluation.

Bank valuations respond to sustainable ROE

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The same book equity supports very different market values when sustainable returns differ.

Indonesian bank case: the lowest P/BV is not enough

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Fictional bank	ROE	Observed P/BV	Model P/BV
Bank A	20%	2.10×	2.00×
Bank B	16%	1.20×	1.50×
Bank C	8%	0.70×	0.50×

Assumed for all banks: cost of equity 12% and sustainable growth 4%.

Bank B trades 20% below its model benchmark.
Bank C is cheapest on P/BV but above its benchmark.

Different credit risk, funding quality or expected loan losses could change this comparison.

Bank and digital-company book values need context

Banks

P/BV is useful when book equity and sustainable ROE capture core economics.

Credit and funding quality

Loan-loss provisions, capital needs and funding costs affect both equity and future ROE.

Digital businesses

Expensed software development and customer acquisition may understate the economic asset base.

Negative or distorted equity

Buybacks, accumulated losses and goodwill can make P/BV difficult to interpret.

EV/Sales links revenue to operating value

EV/Sales = Operating enterprise value / Revenue

Revenue multiples remain measurable when operating earnings are negative, but value still requires future cash generation.

Comparison	Interpretation
EV/Sales	Operating value relative to operating revenue
P/Sales	Equity value relative to total revenue
Main analytical issue	Margins, reinvestment, risk and growth duration

P/S mixes an equity numerator with a pre-financing denominator. Leverage must be controlled.

Margins and capital efficiency determine EV/Sales

$$\text{EV / Sales}_1 = \text{After-tax operating margin} \times (1 - \text{Reinvestment rate}) / (\text{WACC} - g)$$

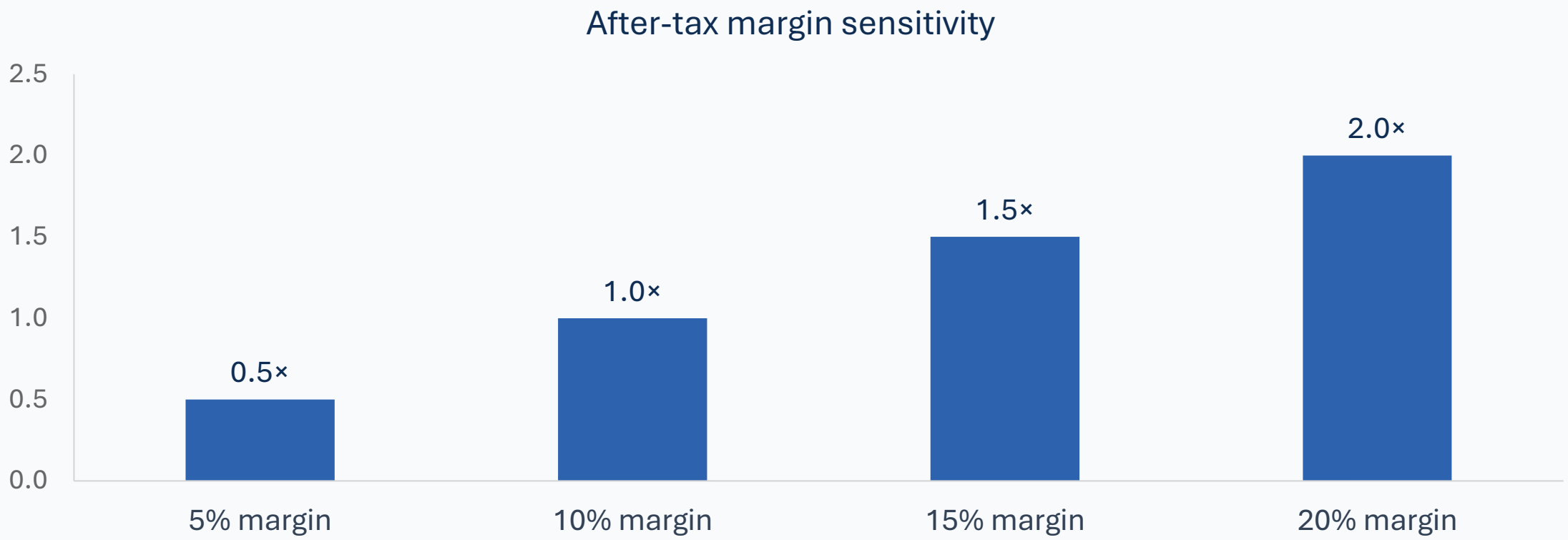
With sustainable growth: Reinvestment rate = g / ROIC.

$$\text{ROIC} = \text{After-tax operating margin} \times \text{Capital turnover}$$

A revenue multiple is a compact forecast of future margins and the investment needed to earn them.

A margin assumption can double the sales multiple

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Illustrative stable growth: reinvestment 40%, WACC 10%, g 4%. Capital turnover adjusts implicitly.

GMV, revenue and gross profit are different bases

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Measure	IDR trillion	Implied multiple
Gross merchandise value (GMV)	100	EV/GMV = 0.20×
Recognized revenue	10	EV/Revenue = 2.00×
Gross profit	4	EV/Gross profit = 5.00×
Enterprise value	20	Same valuation in every row

Illustrative 10% take rate. Merchant sales do not automatically equal platform revenue.

Digital-platform peer selection

Dimension	Why the comparison can fail
Principal versus agent revenue	Gross and net recognition can create very different reported revenue
Take rate and incentives	Monetization and subsidies alter revenue quality and margins
Business mix	Marketplace, logistics, lending and advertising have different economics
Unit economics	Retention, contribution margin and acquisition cost affect future cash flow
Funding and dilution	Cash burn, financing needs and share-based awards affect equity value

Operating metrics explain valuation assumptions. They do not replace a path to cash flow.

Indonesian platform case: sensitivity to the multiple

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Scenario	Low	Base	High
Assumed EV/Revenue	1.5×	2.0×	2.5×
Revenue, IDR trillion	10	10	10
Enterprise value, IDR trillion	15	20	25
Equity value: EV − 2 + 5	18	23	28
Value per share, IDR	900	1,150	1,400

Debt IDR 2tn, cash IDR 5tn, 20bn shares. No NCI or preferred equity.

These are assumed scenarios. Margin potential, cash burn and peer quality must justify the range.

Brand value is already present in the drivers

Pricing power

A brand can support higher margins or reduce the discount needed to retain customers.

Growth and persistence

Customer loyalty may extend profitable growth and protect returns on invested capital.

Valuation treatment

Capture these benefits in the margin, growth, reinvestment and risk assumptions.

Double-counting risk

Adding a separate brand premium after using brand-supported economics can count the same benefit twice.

INTERPRETATION

Market regimes and acquisition multiples

Market pricing changes

Interest rates, risk appetite and earnings expectations can move an entire peer group.

Historical averages need context

A company’s past multiple reflects its past economics and the market conditions of that period.

Trading and transaction multiples

Acquisition prices may include control or synergies that public minority prices do not.

Cross-country comparisons

Currency, inflation, country risk, liquidity and business mix can justify different multiples.

A multiple-based terminal value carries market assumptions

Terminal EV_n = Exit multiple × Matching terminal metric

A forward exit EV/EBITDA at year n applies to EBITDA in year n+1. Discount terminal EV at WACC in an enterprise DCF.

DCF with an exit multiple combines intrinsic forecasts with a relative-pricing terminal assumption.

The terminal multiple must fit mature margins, growth, returns and risk.

CLASS EXERCISE

MBA workshop: a consumer business valuation

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Given inputs	Value
Normalized forward net income / EBITDA	500 / 900
Debt / cash / NCI / preferred equity	2,000 / 300 / 0 / 0
Current shares / quoted price	5 billion / IDR 1,300
Selected forward P/E range	14–16×
Selected forward EV/EBITDA range	9–11×

Company financial amounts in IDR billion, except shares and price.

Tasks: estimate both equity ranges, compare the quoted price, and identify the assumptions that could overturn the conclusion.

CLASS EXERCISE

Workshop solution: two conditional valuation ranges

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Method	Calculation	Value per share
P/E	500 × 14–16 = equity 7,000–8,000	IDR 1,400–1,600
EV/EBITDA	900 × 9–11 = EV 8,100–9,900	
Equity bridge	EV − 2,000 + 300 = 6,400–8,200	IDR 1,280–1,640
At IDR 1,300	Observed P/E 13.0× and EV/EBITDA 9.11×	Mixed evidence

Enterprise and equity values in IDR billion. Shares: 5 billion.

The quote is below the P/E range but inside the EV/EBITDA range. No automatic buy conclusion follows.

JUDGMENT

Reconciling valuation evidence

Select a primary method

Choose the multiple with the strongest link to the target’s economics and best-quality data.

Explain disagreement

Check leverage, depreciation, taxes, forecast periods and normalization before averaging results.

Cross-check with DCF

Compare implied growth, margins and reinvestment with economically sustainable assumptions.

Communicate a conditional range

State the benchmark, key assumptions and evidence that would change your judgment.

Relative valuation review checklist

Question	Evidence needed
Is the multiple internally consistent?	Claimholders, consolidation, leases and cash treatment
Are target and peer data comparable?	Same date, horizon, currency basis and adjustment rules
Does the benchmark reflect fundamentals?	Growth, risk, ROE/ROIC, margins and reinvestment
How robust is the range?	Peer inclusion, outliers and scenario sensitivity
Does enterprise value reconcile to common equity?	Debt, cash, NCI, preferred claims and dilution
What could invalidate the conclusion?	Earnings quality, financing needs or market-wide mispricing

Source guide and further reading

Supplied material	Coverage in this lecture
Session 14: Relative valuation basics	Definitions, four tests, peers and distribution
Session 15: PE ratios	P/E variants, fundamentals and earnings quality
Session 16: Other Earnings Multiples	Enterprise multiples and cash conversion
Session 17: PBV ratios	Book value, ROE and banking applications
Session 18: Revenue Multiples	Margins, reinvestment and brand economics

Damodaran, Investment Valuation, 3rd ed., Chapters 17–20.
NYU Stern companion materials: pages.stern.nyu.edu/~adamodar/

All Indonesian company cases are fictional teaching examples. Historical source data are not current quotes.

REFERENCES

Damodaran, A. (2012). Investment Valuation: Tools and Techniques for Determining the Value of Any Asset, 3rd Edition (University Edition). Hoboken, NJ: John Wiley & Sons. Chapters 17-20

Companion datasets and valuation spreadsheets: damodaran.com

Thank You

Questions & Discussion

